

Vulnerability Assessment and Penetration Testing (VAPT)

Report – Capstone Project (Week 4)

Prepared By: Rahul Rajput

Course: Cyber Security Internship – CYART

Submission Date: 27 February 2026

Tools Used: Kali Linux, Metasploit, Burp Suite, MobSF, LinPEAS

Project Overview:

This report presents the findings of a full Vulnerability Assessment and Penetration Testing (VAPT) engagement conducted in a controlled lab environment. The objective was to identify security weaknesses, exploit them ethically, and provide remediation recommendations to improve overall security posture.

1. Introduction

This report documents the Vulnerability Assessment and Penetration Testing (VAPT) conducted in a controlled lab environment. The assessment focused on identifying vulnerabilities in Metasploitable2 and the InsecureBankv2 Android application. The objective was to perform ethical exploitation, analyze security weaknesses, and provide remediation recommendations.

2. Scope of Assessment

Targets:

Metasploitable2 (Linux vulnerable VM)

DVWA (Web Application)

InsecureBankv2 (Android APK)

Tools Used:

Kali Linux

Metasploit

Burp Suite

MobSF

LinPEAS

3. Methodology

1. Reconnaissance
2. Scanning & Enumeration
3. Exploitation
4. Privilege Escalation
5. Reporting

4. Exploitation Phase – Metasploitable2

4.1 VSFTPD 2.3.4 Backdoor Exploitation

The target system was found running VSFTPD 2.3.4 which contains a known backdoor vulnerability.

Metasploit module used:

exploit/unix/ftp/vsftpd_234_backdoor

Successful exploitation resulted in shell access.

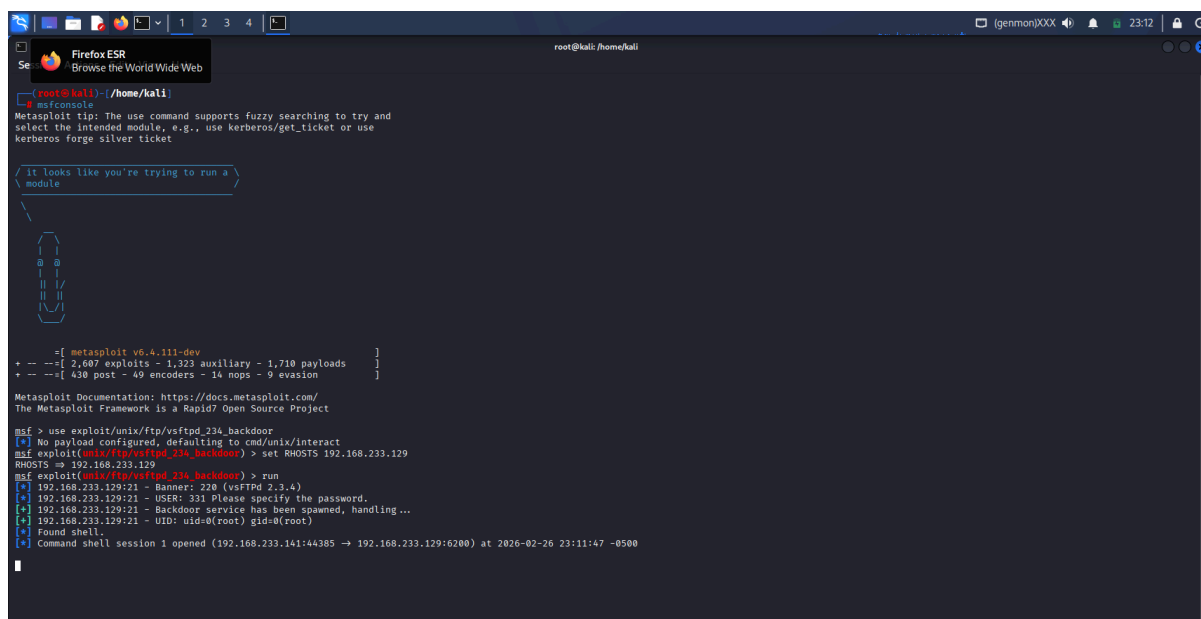


Figure 1: Successful exploitation of VSFTPD 2.3.4 using Metasploit resulting in command shell access.

5. Privilege Escalation

LinPEAS was executed to perform local enumeration on the target system.

The scan identified that the system is running an outdated Linux kernel (2.6.24-16-server), which indicates potential privilege escalation vulnerabilities.

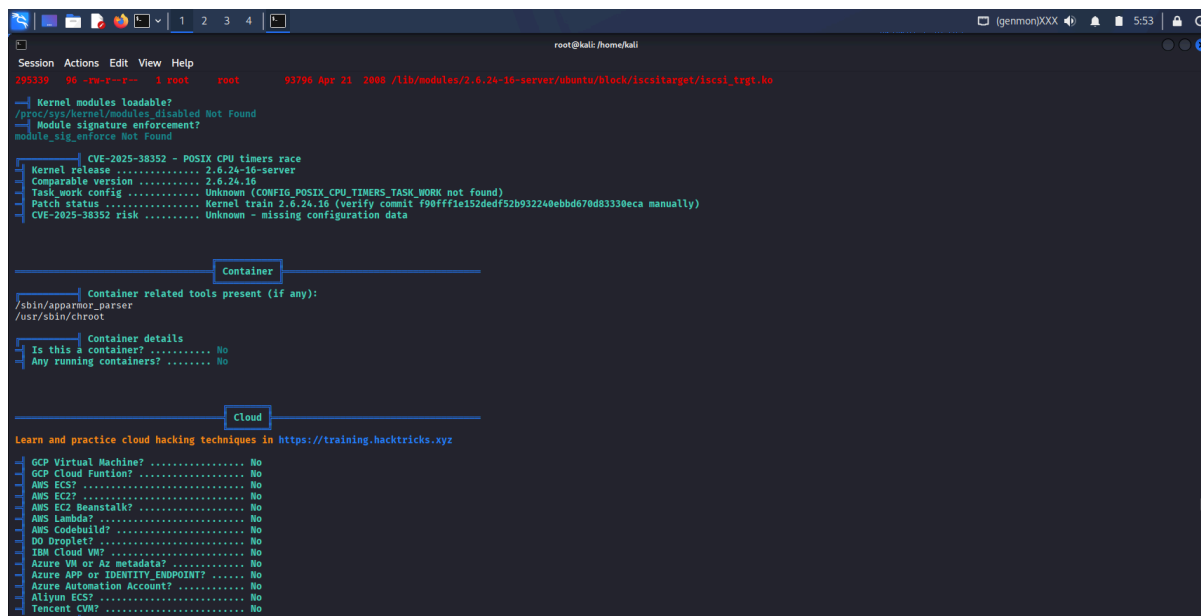


Figure 2: LinPEAS output showing outdated kernel version indicating potential privilege escalation vulnerability.

6. Web Application Testing – DVWA

6.1 SQL Injection Testing

SQL Injection was tested on DVWA using payload:

1' OR 1=1#

The application returned unauthorized results, confirming vulnerability.

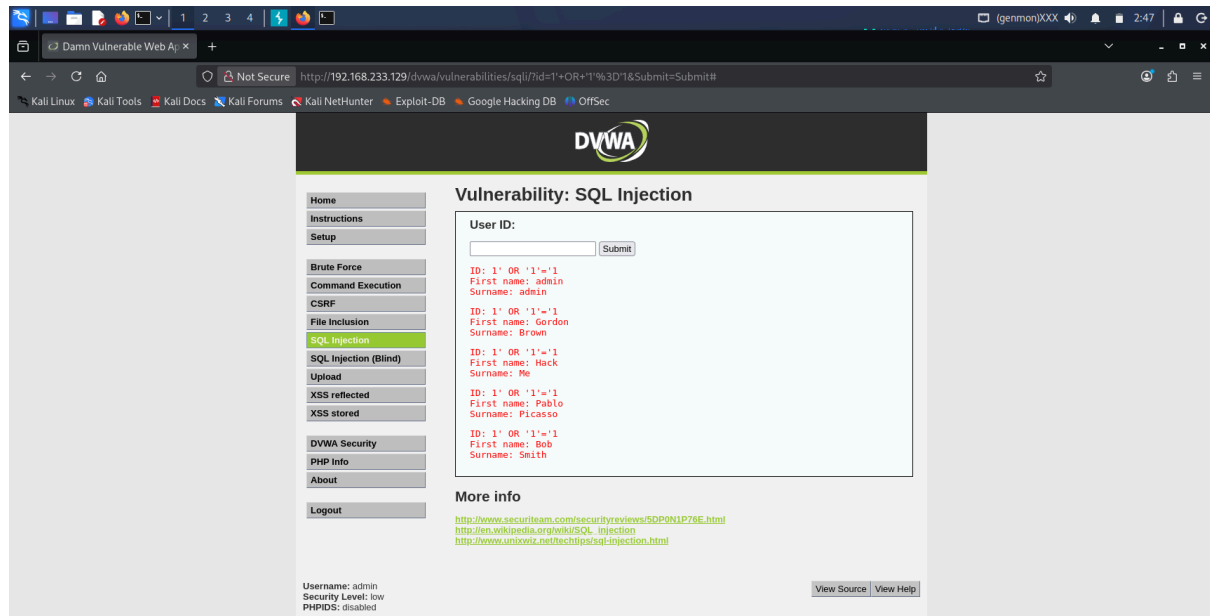


Figure 3: SQL Injection payload bypassing input validation and retrieving multiple user records in DVWA.

7. Mobile Application Testing – InsecureBankv2

Static analysis of the InsecureBankv2 Android application was performed using MobSF (Mobile Security Framework).

7.1 Security Score Analysis

MobSF generated a low security score, indicating multiple high-risk findings within the application. The analysis highlighted insecure configurations and vulnerable components that increase the attack surface.

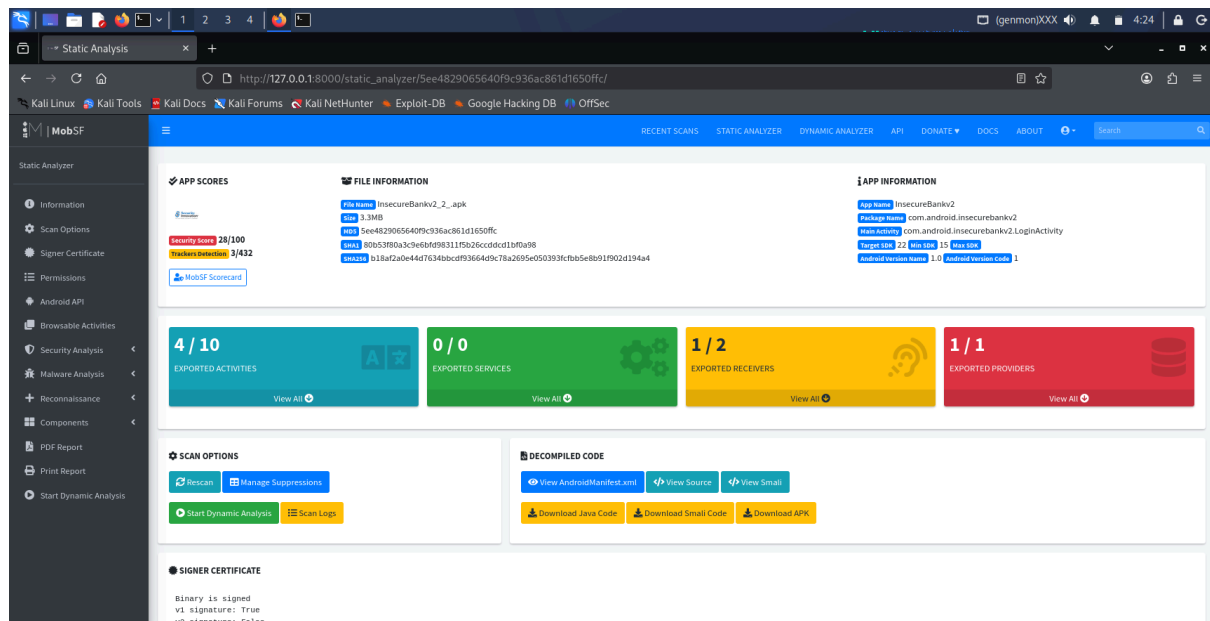


Figure 4: MobSF Security Scorecard showing high-risk findings in the application.

7.2 Dangerous Permissions Identified

The AndroidManifest.xml analysis revealed several dangerous permissions that could potentially compromise user privacy and security. These include:

- READ_CONTACTS
- SEND_SMS
- READ_PROFILE

These permissions, if misused, could allow attackers to access sensitive user data or perform unauthorized actions.

- Apply least privilege principle for permissions.
- Regularly perform security audits and patch updates.

10. Executive Summary

This assessment identified critical security vulnerabilities across the tested systems. The Metasploitable2 server was vulnerable to remote code execution, allowing attackers to gain shell access. The DVWA web application was susceptible to SQL Injection, enabling unauthorized database access. The InsecureBankv2 mobile application showed insecure storage, dangerous permissions, and hardcoded secrets. If deployed in a real-world environment, these vulnerabilities could result in data breaches and system compromise. Immediate patching, secure coding practices, and continuous security assessments are strongly recommended to enhance the overall security posture.